

5.1.16 Factorial and "n Choose r" Notation

An important use for the factorial notation is in calculating values of quantities, called n choose r , that occurs in many branches of mathematics, especially those connected with the study of counting techniques and probability.

Definition

Let n and r be integers with $0 \leq r \leq n$. The symbol

$$\binom{n}{r}$$

is read "**n choose r**" and represents the number of subsets of size r that can be chosen from a set with n elements.

Observe that the definition implies that $\binom{n}{r}$ will always be an integer because it is a number of subsets. In Section 9.5 we will explore many uses of n choose r for solving problems involving counting, and we will prove the following computational formula:

Formula for Computing $\binom{n}{r}$

For all integers n and r with $0 \leq r \leq n$,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$